AY2024/25 - Semester 1

CM3191

Chemical Synthesis Experiments

(INORGANIC Chemistry)



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SCHEDULE

Date*	Week 2	Week 3	Week 4	Week 5	Week 6
Inorganic practical	Experiment **1** (Session I)	Experiment **1** (Session II)	Experiment **1** (Session III)	Experiment **2** (Session I)	Experiment **2** (Session II)

**Remarks:*

(i) The first inorganic lecture will be held during the lecture slots in Week 1.

(ii) The date(s) and the schedule(s) of experiment(s) will be different for the makeup case. Please contact Dr. Chan directly for inorganic experiments.

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

A. INTRODUCTION

Homogenous Catalyst is a catalyst dissolving in the reaction mixture, and often works under milder conditions than heterogeneous catalyst. Tris(triphenylphosphine)ruthenium(II) dichloride, $RuCl_2(PPh_3)_3$, is one of the non-organometallic homogenous catalysts in transition-metal-catalyzed oxidations of alcohols.[1] Besides, $RuCl_2(PPh_3)_3$ functions as a precatalyst in various reactions, such as C-H Activation in a Ruthenium/Lewis Acid System and the synthesis of furans from allenyl sulfides.[2] It can be also used as a precursor in the preparation of organometallic ruthenium dihydride complex and Gubbs' catalyst.[3]

Wilkinson and Stephenson's research groups reported in 1966 that with different reaction conditions, the interaction of triphenylphosphine (PPh_3) with hydrated ruthenium trichloride, $RuCl_3 \cdot xH_2O$, leads to a variety of mononuclear complexes such as $RuCl_2(PPh_3)_4$, $RuCl_2(PPh_3)_3$, and $RuCl_2(PPh_3)_3(CH_3OH)$.[4]

Interestingly, Placa and Ibers published the X-ray structure of $RuCl_2(PPh_3)_3$, which was prepared and studied by Vaska.[5] It was found that Ru(II) in $RuCl_2(PPh_3)_3$ is five-coordinated, and $RuCl_2(PPh_3)_3$ is a diamagetic and monomeric compound. Such X-ray results also showed Ru lies toward the center of a distorted square pyramid containing *trans* chlorine atoms and *trans* phosphorus atoms in the base and an apical phosphorus atom as depicted in **Figure 1**.

repelled by magnetic field as all e^- are paired. no unpaired e^- to align w external magnetic field

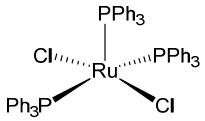


Figure 1. Molecular structure of $RuCl_2(PPh_3)_3$.[4]

The oxidation of alcohols to carbonyl compounds is an important physiological transformation and fundamental reaction in organic synthesis. In human body, ethanol is broken down in the liver by an enzyme named alcohol dehydrogenase (ADH), which is a class of zinc enzymes, into acetaldehyde. The acetaldehyde is carcinogenic and it is then quickly converted to acetate by another enzyme called aldehyde dehydrogenase (ALDH) and such acetate is finally metabolized into water and carbon dioxide for easy elimination.[6] In organic chemistry, there are several methods and reagents to carry out the oxidation of alcohols.[7] The use of catalytic reactions in such oxidations is important in a large-scale process in industry.[8]

In this experiment, $RuCl_2(PPh_3)_3$, is prepared and used as a homogenous precatalyst to convert the alcohol (1-phenylethanol) into ketone (acetophenone) *via* a homogeneous hydrogen-transfer oxidation reaction, in which acetone is employed as solvent and hydrogen acceptor. Bäckvall *et al.* proposed a mechanism in 1996 for the $RuCl_2(PPh_3)_3$-catalyzed oxidation of 1-phenylethanol to acetophenone (**Figure 2**).[1,9] Later, in 1999, Szabó and Bäckvall *et al.* proposed another reaction mechanism involving ruthenium dihydride, $RuH_2(PPh_3)_3$ as an active catalyst in such catalytic reaction (**Figure 3**).[10] Both mechanisms involve β-hydride elimination from ruthenium alkoxide (**Figure 4**).

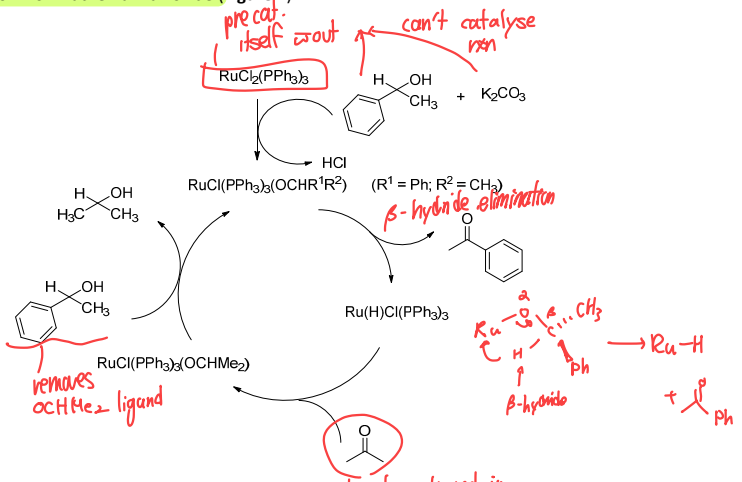


Figure 2. Proposed mechanism for the $RuCl_2(PPh_3)_3$-catalyzed oxidation of 1-phenylethanol to acetophenone by Bäckvall *et al.* in 1996.[1,9]

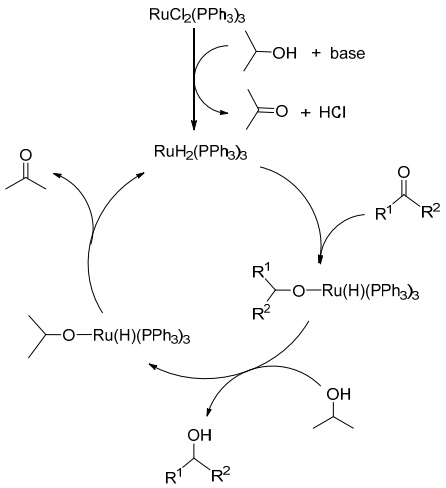


Figure 3. Proposed mechanism for the $RuCl_2(PPh_3)_3$-catalyzed oxidation of 1-phenylethanol to acetophenone by Szabó and Bäckvall *et al.* in 1999.[10]

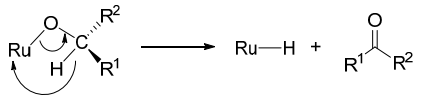


Figure 4. β-hydride elimination involved in the $RuCl_2(PPh_3)_3$-catalyzed oxidation of 1-phenylethanol to acetophenone.[1]

$RuCl_2(PPh_3)_3$ is sensitive to moisture and solvent.[5] The moisture-free reflux setup, in which a water-cooled reflux condenser is connected with a calcium chloride drying tube, is used in this experiment (**Figure 5**) for both synthesis of $RuCl_2(PPh_3)_3$ and its catalytic reaction, and the solvents used must be dried with molecular sieves.

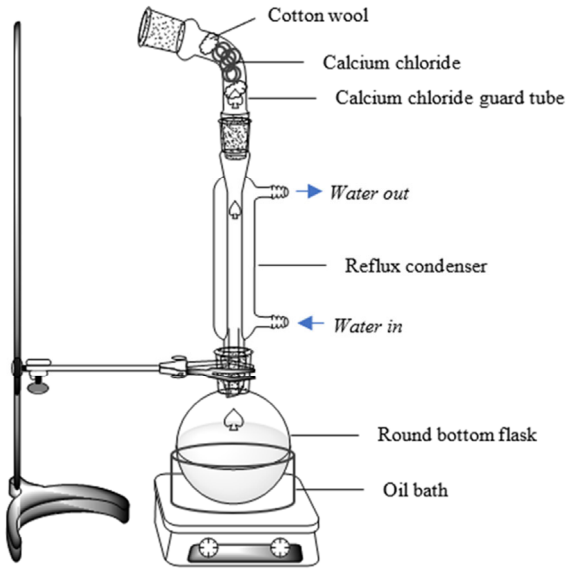


Figure 5. Apparatus used in the synthesis of $RuCl_2(PPh_3)_3$ and its catalytic studies for this experiment.

This is an individual experiment. During the hands-on sessions, you are required to firstly prepare $RuCl_2(PPh_3)_3$ from $RuCl_3 \cdot xH_2O$, of which the attenuated total reflectance (ATR) IR spectrum is depicted in **Figure 6**, and to do the respective characterization. The catalytic oxidation of 1-phenylethanol to acetophenone in the absence and presence of $RuCl_2(PPh_3)_3$ will be studied in Session II and III respectively with a specific experimental condition. Due to the moisture-sensitivity of $RuCl_2(PPh_3)_3$, the preparation of $RuCl_2(PPh_3)_3$ and its catalytic reaction will be performed using dried solvents and calcium chloride guard tube in the reflux setup.

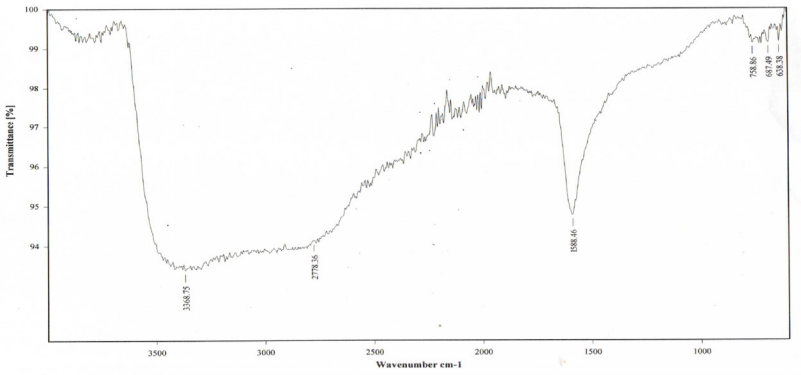


Figure 6. ATR-IR spectrum (solid) of $RuCl_3 \cdot xH_2O$ used.

Safety Precautions

Ruthenium(III) chloride hydrate ($RuCl_3 \cdot xH_2O$) is corrosive. Tris(triphenylphosphine)ruthenium(II) dichloride [$RuCl_2(PPh_3)_3$], 1-phenylethanol, acetophenone, mesitylene, potassium carbonate (K_2CO_3), Celite, potassium bromide (KBr), calcium chloride ($CaCl_2$) and reaction products are all harmful. Organic reagents and solvents such as mesitylene, ethanol, acetone and hexane are highly flammable. Triphenylphosphine (PPh_3), deuterated chloroform ($CDCl_3$) and hexane are toxic and irritating. Do not inhale them or allow them to contact the skin. Handle them with appropriate personal protective equipment (PPE). Perform all operations in a well-ventilated fume cupboard. Refer to the MSDS of chemicals used for detailed safety information.

Disposal of Chemical Wastes

Heavy metals (i.e. ruthenium waste) should be disposed of in heavy metal waste container. Chlorinated and non-chlorinated solvents should be disposed of in separate halogenated and non-halogenated containers respectively. The Celite must be discarded into the designated container in the designated fume cupboard after use. If unsure about the disposal procedure, please check with lecturer/lab officer/graduate teaching assistants (GTAs).

B. EXPERIMENTAL

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

1. Weigh 0.50 g of PPh_3 and transfer it into a 100 mL round bottom flask fitted with a stirrer. Then add 20 mL of dried absolute ethanol into the flask. Warm the solution (~ 40 °C) for 10 - 15 minutes to allow the solids to dissolve.

2. Weigh 0.08 g of $RuCl_3.xH_2O$ and transfer it into a 50 mL round bottom flask fitted with a stirrer bar. Then add 20 mL of dried absolute ethanol into the flask. Allow the $RuCl_3.xH_2O$ solid to dissolve *via* constant stirring using the stirrer bar.

3. Using a glass dropper, transfer the ethanolic solution containing the dissolved $RuCl_3.xH_2O$ into the 100 mL round bottom flask containing the dissolved PPh_3 in ethanol. Add 5 mL of dried absolute ethanol to rinse all $RuCl_3.xH_2O$ residue into the reaction flask containing PPh_3.

4. Connect a water-cooled reflux condenser with a calcium chloride drying tube to the 100 mL round bottom flask containing the reactants [*See **Figure 5***].

5. Reflux the reaction mixture for **90 minutes**. *[Note: the temperature of oil bath is < 130 °C.]*

6. Allow the reaction mixture to cool to room temperature after the reflux.

7. Using a Hirsch funnel, collect the solid, $RuCl_2(PPh_3)_3$ by suction filtration. Wash the solid with cold dried absolute ethanol (2 x 3 mL) and then dry it under vacuum for 20 – 30 minutes.

8. Weigh the amount of dried $RuCl_2(PPh_3)_3$ obtained and record it on datasheet.
   [Note: The detailed calculation of % yield of $RuCl_2(PPh_3)_3$ obtained must be shown in the proforma.]

9. Transfer the dried $RuCl_2(PPh_3)_3$ obtained (~ 10 mg) into a sample tube. Label the sample tube clearly and submit it for Electrospray Ionisation Mass Spectrometry (ESI-MS) analysis.

10. Obtain the solid IR spectrum (range: 500–4000 cm^{-1}) of the dried $RuCl_2(PPh_3)_3$ obtained by *either* ATR-IR *or* KBr disc.
    [Note: The IR characterization method chosen must be recorded on your datasheet.]

Session II: Controlled Study - Oxidation of 1-phenylethanol to acetophenone in the absence of Catalyst

1. Add 8 mL of dried acetone and a magnetic stirrer bar into a 100 mL round bottom flask.
2. Using a micropipette, add 0.06 mL of 1-phenylethanol into the reaction flask upon stirring.
3. Upon stirring, add 0.07 g of K_2CO_3 into the reaction flask.
4. Connect quickly a water-cooled reflux condenser with a calcium chloride drying tube to the reaction flask containing the reactants [*See* ***Figure 5***].
5. Reflux the reaction mixture for **60 minutes**. *[Note: The temperature of oil bath is < 100 °C.]*
6. Cool the reaction mixture to room temperature after the reflux.
7. Upon stirring, add 50 mL of *n*-hexane into the reaction flask to quench the reaction.
8. Using a Hirsch funnel, filter the reaction mixture through Celite by suction filtration and collect the filtrate. *[Note: The suction flask must be washed and dried completely before use.* ***When the suction is on, a thin layer of Celite should be packed on top of the filter paper in the Hirsch funnel****.]*
9. Transfer the filtrate from the suction flask into a 150 mL round bottom flask. Rinse the suction flask with *n*-hexane (2 x 5 mL) and transfer it into the round bottom flask.
10. Using a rotary evaporator, remove the solvent from the reaction mixture in *n*-hexane under reduced pressure.
11. Connect the round bottom flask with a gas adapter and dry the oily product under vacuum for 15 - 20 minutes.
12. Using a micropipette, add 0.07 mL of mesitylene into the reaction flask upon stirring.
13. Add 0.6 mL of $CDCl_3$ into the round bottom flask containing oily product and transfer all solution into an NMR tube with a clean glass dropper. Label the NMR tube clearly and submit it for ^1H NMR analysis.

Session III: Study of Catalytic Reaction: $RuCl_2(PPh_3)_3$-catalysed Oxidation of 1-phenylethanol to acetophenone

1. Add 8 mL of dried acetone and a magnetic stirrer bar into a 100 mL round bottom flask.
2. Using a micropipette, add 0.06 mL of 1-phenylethanol into the reaction flask upon stirring.
3. Upon stirring, add 0.07 g of K_2CO_3 into the reaction flask.
4. Lastly, add 0.06 g of your $RuCl_2(PPh_3)_3$ obtained into the reaction flask upon stirring, and then quickly connect a water-cooled reflux condenser with a calcium chloride drying tube to the reaction flask containing the reactants [*See* ***Figure 5***].
5. Reflux the reaction mixture for **60 minutes**. *[Note: The temperature of oil bath is < 100 °C.]*
6. Cool the reaction mixture to room temperature after the reflux.
7. Upon stirring, add 50 mL of *n*-hexane into the reaction flask to quench the reaction.
8. Using a Hirsch funnel, filter the reaction mixture through Celite by suction filtration and collect the filtrate. *[Note: The suction flask must be washed and dried completely before use.* ***When the suction is on, a thin layer of Celite should be packed on top of the filter paper in the Hirsch funnel****.]*
9. Transfer the filtrate from the suction flask into a 150 mL round bottom flask. Rinse the suction flask with *n*-hexane (2 x 5 mL) and transfer it into the round bottom flask.
10. Using a rotary evaporator, remove the solvent from the reaction mixture in *n*-hexane under reduced pressure.
11. Connect the round bottom flask with a gas adapter and dry the oily product under vacuum for 15 - 20 minutes.
12. Using a micropipette, add 0.07 mL of mesitylene into the reaction flask upon stirring.
13. Add 0.6 mL of $CDCl_3$ into the round bottom flask containing oily product and transfer all solution into an NMR tube with a clean glass dropper. Label the NMR tube clearly and submit it for ^1H NMR analysis.

2. LAB REPORT

Individual ***softcopy*** *of* ***handwritten*** *proforma lab report* (*via* Canvas-Assignments) consisting of all of your own datasheet(s) and spectra obtained which must be signed by GTA(s) at the end of each practical session. Refer to CM3191-inorganic lecture notes for the requirement of the proforma lab report. The "ONE-file" softcopy proforma in pdf format (attached with datasheets, all spectra and appendices, if any) must be uploaded into the folder named "Inorganic Expt. 1 Proforma submission" *via* Canvas-Assignments. Hardcopy of the proforma lab report will NOT be accepted. Penalty will be applied for any late submission.

3. REFERENCES

1. Almeida, M. L. S.; Bellar, M.; Wang, G.-Z.; Bäckvall, J.-E., *Chem. Eur. J*. **1996**, 2, *1533*.
2. (a) Zhang, S. –Y.; Tu, Y. –Q.; Fan, C. –A.; Jiang, Y. –J.; Shi, L.; Cao, K.; Zhang, E., *Chem. Eur. J*. **2008**, 14, *10201*. (b) Peng, L.; Zhang, X.; Ma, M.; Wang, J., *Angew. Chem. Int. Ed. Engl.*, **2007**, 46, *1905*.
3. (a) Samouei, H.; Grushin, V.V, *J. Organometallics*, **2013**, 32, *4440*. (b) Schwab, P.; France, M. B.; Ziller, J. W. and Grubbs, R. H. *Angew. Chem. Int. Ed. Engl.*, **1995**, *34*, 2039-2041.
4. (a) Stephenson, T. A.; Wilkinson G., *J. Inorg. Nucl. Chem*., **1966**, 28, *945*. (b) Hallman, P. S.; Stephenson, T. A.; Wilkinson G., *Inorg. Synth.*, **1970,** 12, *238*.
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7. Haines, A. H., *Methods for the Oxidation of Organic Compounds*, Academic Press, London, **1988**.
8. Sheldon, R. A.; Kochi, J. K., *Metal-Catalyzed Oxidation of Organic Compounds*, Academic Press, New York, **1981**.
9. Miecznikowski, J. R.; Caradonna, J. P.; Foley, K. M.; Kwiecien, D. J.; Lisi, G. P.; Martinez, A. M., *J. Chem. Educ.* **2011**, 88, *657*.
10. Aranyos, A.; Csjernyik, G.; Szabó, K. J.; Bäckvall, J. -E., *Chem. Commun*., **1999**, 351-352.

4. APPENDICES *(available in Canvas-Files for inorganic experiments)*

Material Safety Data Sheets (MSDSs) of Ruthenium(III) chloride hydrate ($RuCl_3 \bullet xH_2O$), triphenylphosphine (PPh_3), tris(triphenylphosphine)ruthenium(II) dichloride [$RuCl_2(PPh_3)_3$], 1-phenylethanol, acetophenone, triphenylphosphine oxide, mesitylene, potassium carbonate (K_2CO_3), Celite, mesitylene, ethanol, acetone, hexane and deuterated chloroform ($CDCl_3$).

INORGANIC Experiment 2

Metallocene: Ferrocene - Synthesis and Characterization

A. INTRODUCTION

The synthesis and basic characterization of a new organoiron compound, ferrocene were reported about seven decades ago by two research groups, T. J. Kealy and P. L. Pauson from Duquesne University in the United States and S. A. Miller, J. A. Tebboth and J. F. Tremaine from the British Oxygen Company, Ltd., in the United Kingdom.[1] However, the correct sandwich structure, [$Fe(\eta^5\text{-}C_5H_5)_2$], was reported in 1952 by G. Wilkinson, M. Rosenblum, M. C. Whiting and R. B. Woodward.[2] The independent pioneering work in the chemistry of ferrocene led to the joint award of the Nobel Prize in Chemistry to both G. Wilkinson and E. O. Fischer in 1973.[3]

As a typical metallocene, ferrocene can exist in two different configurations: eclipsed (D_{5h} symmetry) and staggered (D_{5d} symmetry) (**Figure 1**). Such two conformers can be differentiated by a small energy gap.[4] It also has been stated in literatures that a staggered conformer dominates experiments in the condensed phase whereas the eclipsed structure is found in the gas phase. Since most of the theoretical studies of ferrocene refer to gas phase or solutions, the eclipsed structure of ferrocene has been more extensively studied relatively.[4]

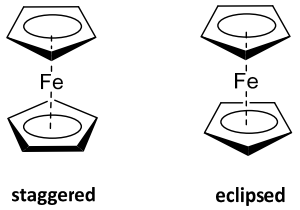


Figure 1. Staggered and eclipsed configurations of ferrocene.

Experimentally, it was found that ferrocene is a stable compound that the complex obeys the 18-electron rule. The metal center in ferrocene was found to be protected by the two cyclopentadienyl ligands.[1] Such discovery has initiated the studies on cyclopentadienyl (Cp) and its derivatives and other delocalized hydrocarbon metal complexes with a unique sandwich molecular structure, strong metal-ring π bonding, and facile redox behavior. A wide range of application for ferrocene and its derivatives has been studied in medicinal chemistry, homogeneous catalysis and materials science.[5]

This is an individual experiment. During the hands-on sessions, you are required to firstly synthesize the crude ferrocene from potassium cyclopentadienide prepared and iron(II) chloride tetrahydrate and to purify and characterize it as listed in the experimental procedures. As the freshly prepared cyclopentadiene monomer and potassium cyclopentadienide prepared are moisture- and air-sensitive, the synthesis of ferrocene will take place under nitrogen atmosphere.

Safety Precautions

Cyclopentadiene is colorless harmful, irritating, smelly liquids. 1,2-dimethoxyethane, acetonitrile, cyclopentadiene, ferrocene and methanol are flammable. 1,2-dimethoxyethane, acetonitrile, calcium chloride, deuterochloroform, dimethyl sulfoxide, ferrocene, potassium cyclopentadienide and potassium bromide are harmful and irritating to skin and eyes. 1,2-dimethoxyethane, cyclopentadiene, hydrochloric acid, iron(II) chloride tetrahydrate and potassium hydroxide are toxic and corrosive. Do not inhale them or allow them to contact skin and eyes. Do handle all chemicals and perform all operations in a well-ventilated fume cupboard. Handle them with appropriate personal protective equipment (PPE) such as safety goggle, lab coat and gloves. Refer to the MSDS of chemicals used for detailed safety information.

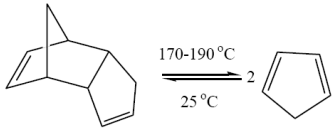
Disposal of Chemical Wastes

Chlorinated and non-chlorinated solvents should be disposed of in separate halogenated and non-halogenated containers respectively. All aqueous waste solutions containing metal ions should be poured into the designated aqueous metal waste solution container. If unsure about the disposal procedure, please check with lecturer/lab officer/graduate teaching assistants (GTAs).

B. EXPERIMENTAL AND ANALYSIS

Session 0: "Cracking" of dicyclopentadiene [prepared by lab officers]

The cyclopentadiene monomer is not stable and has to be obtained from the cracking of the dimer, dicyclopentadiene (4,7-methylene-4,7,8,9-tetrahydroindene) (**Equation 1**).



dicyclopentadiene
density: 0.98 g/mL
b.p.: 170 °C
M.W.: 132.20

cyclopentadiene (1,3-cyclopentadiene)
density: 0.80 g/mL
b.p.: 41 °C
M.W.: 66.10

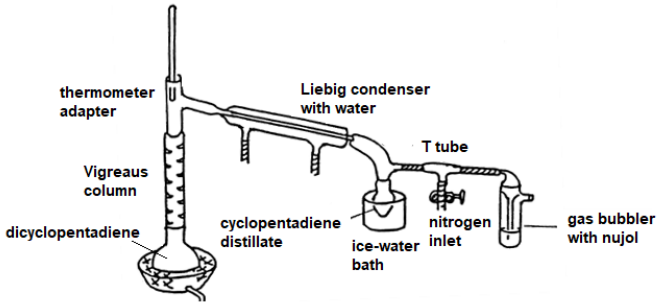


Figure 2. Apparatus used in the cracking of dicyclopentadiene.

Add 75 mL of commercial dicyclopentadiene to the 250 mL distilling flask (**Figure 2**). First connect the heating mantle, and turn on the water flow to the condenser. Check that all joints are well sealed and that the apparatus is clamped tightly. Insert a syringe needle into the septum of the two-neck round-bottom flask (as receiver flask), flush the system with nitrogen gas for 1 minute, and then remove the vent syringe needle.

Start the cracking process by heating the distillation flask until it is hot (temperature setting 80-90 °C), and then reduce the heating (temperature setting ~ 50 °C) to avoid "flooding" the Vigreux column. [Note: During this time, the liquid in the flask will start to bubble and the plates in the Vigreux column will become wet with condensation, indicating that the cracking process is taking place.] The temperature at the top of the column should rise to 39 °C and the condensation of the vapor should begin in the condenser.

Collect cyclopentadiene monomer boiling at 39-41 °C. [Note: Maintain a rate of distillation that does not exceed 2-3 drops/second of monomer passing from the water condenser to the ice-cooled receiver flask. During this time the rate of cracking may decrease. In this event a periodic increase in the temperature of heating mantle setting may be required.].

Use a syringe to take 1.0 mL of cyclopentadiene monomer for the synthesis of ferrocene in Session I (a). [Note: The monomer must be used immediately or the cracking and distillation procedure will have to be repeated.]

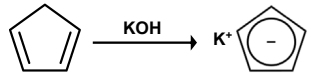
Make sure the two solutions of potassium hydroxide and the iron(II) chloride are ready BEFORE obtaining the monomer.

Session I: Synthesis and Microscopic Image of Crude Ferrocene

I-a. Synthesis of Crude Ferrocene

$$2C_5H_6 + 8KOH + FeCl_2{\cdot}4H_2O \rightarrow Fe(C_5H_5)_2 + KCl + 6KOH{\cdot}H_2O$$ ***Equation 2***

(i) Preparation of potassium cyclopentadienide (also known as cyclopentadienyl) solution



Equation 3

Place a rubber septum in the one of the side necks of a 100 mL three-neck (14/20) round-bottom flask and a glass stopper in another side neck. Quickly weigh and add 4.0 g of KOH pellets into the round-bottom flask via the middle neck, and cap the flask with a calcium chloride guard tube in the middle neck. Quickly add 12 mL of anhydrous 1,2-dimethoxyethane (DME) from the dispenser into the flask and cap the flask with a calcium chloride guard tube.

Upon stirring, quickly replace the guard tube with a glass transfer gas adapter connected to the nitrogen gas (N_2) supply and a bubbler on the top, and replace a glass stopper with a pressure-equalizing dropping funnel (DF) [See **Figure 3**]. Turn on the N_2 supply tap slowly and maintain the inert atmosphere (about 1 bubble of N_2 per second) for about 3-5 minutes before proceeding to the next step.

Upon stirring, use a syringe connected to a needle to add 1.0 ml of cyclopentadiene* (prepared by lab officers in Session 0) into the three-neck round-bottom flask via septum. Stir for 10 minutes to obtain a cyclopentadienide solution.

*[*Warnings: (i) Ensure the solution of iron(II) chloride has been prepared and added into the pressure-equalizing dropping funnel before adding cyclopentadiene monomer into the three-neck round-bottom flask. (ii) Use the needle with care as it is sharp. (iii) Both syringe and needle must be degassed with N_2 first in the receiver flask of the setup of cyclopentadiene, before collecting 1.0 ml of cyclopentadiene. (iv) Inform the lab officer or your GTA or the lecturer before doing such step.]*

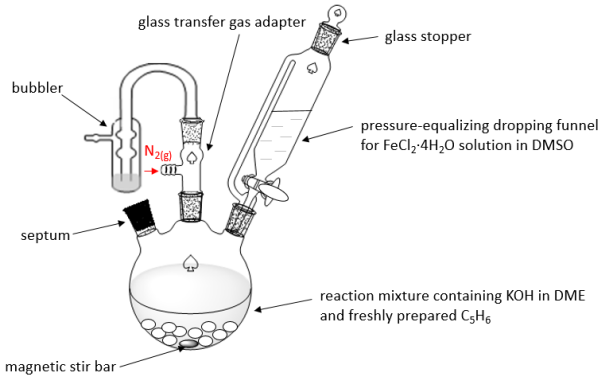


Figure 3. Experimental setup for the synthesis of ferrocene

(ii) Preparation of iron(II) chloride solution

Quickly add 1.3 g of $FeCl_2 \cdot 4H_2O$ in a 25 mL round-bottom flask and cap it with calcium chloride guard tube. Add 7 ml of anhydrous dimethylsulfoxide (DMSO) with stirring. Once $FeCl_2 \cdot 4H_2O$ has dissolved into DMSO, use a glass funnel to quickly transfer the solution to the pressure-equalizing dropping funnel (DF). Add ~3.5 ml of anhydrous DMSO to rinse the round-bottom flask and transfer the solution to DF quickly via a glass funnel.

(iii) Formation of ferrocene

$$2\ KC_5H_5 + FeCl_2 \rightarrow Fe(C_5H_5)_2 + 2\ KCl \qquad \textit{Equation 4}$$

Upon stirring, add the $FeCl_2$ solution slowly (or dropwise at a rate of ~ 20 drops per minute) into the round-bottom flask containing potassium cyclopentadienyl. Continue stirring the mixture for another 30 minutes. Maintain the gentle flow of N_2 in the entire process.

Prepare an ice acidic solution by slow addition^ of 10 mL of concentrated HCl into 30 g of ice in a 250 mL beaker. After 5 minutes, pour 10 mL of the ice acidic solution into a 25 mL beaker. These two beakers of ice acidic solutions will be used in the next step.

[^Warnings: (i) Concentrated HCl is highly corrosive. (ii) Inform the lab officer or your teaching assistant or the lecturer before doing such step.]

For the ferrocene mixture, after 30-minute stirring, stop the N_2 flow. Remove all the glassware connected to the three-neck round-bottom flask. Upon stirring with a glass rod, slowly add the ferrocene reaction mixture from the three-neck round-bottom reaction flask into the 250 mL beaker containing concentrated HCl and ice. Place the three-neck round-bottom reaction flask in an ice-water bath. Add the 10 mL of the ice acidic solution from a 25 mL beaker into the three-neck round-bottom reaction flask upon stirring. Swirl the round-bottom flask and pour all the solution mixture into the 250 mL beaker containing reaction mixture. Rinse the round-bottom flask with 10 mL of deionized water and transfer it into the 250 mL beaker containing reaction mixture. Stir the resulting mixture for 15-20 minutes. Collect the crude ferrocene by suction filtration (with a Büchner funnel). Wash the crude ferrocene with deionized (D.I.) water (3 x 20 mL), and dry it under vacuum for ~30 minutes. Record the amount of crude ferrocene obtained and transfer it into a clean sample vial for the next lab session.

I-b. Microscopic Image of Crude Ferrocene

Record a microscopic image of your crude ferrocene [obtained in Session I (a)] by using the stereomicroscope. With the aid of GTA or lecturer-in-charge, use the lowest magnification ratio of **14x**, select the best view and record the image of your product clearly. Type your matric number on the image captured and send the softcopy of the image to your handphone or NUS email address.

[*Note: The microscopic image obtained must be attached as a part of appendices in the short lab report.*]

Session II: Purification and Characterization of Ferrocene

II-a. Purification of ferrocene

Record the weights of the glass petri dish covers (bigger size) first with an analytical balance. Spread the crude ferrocene on a glass petri dish base (smaller size) and cover it with the glass petri dish cover (bigger size). Purify the crude ferrocene by sublimation using the setup depicted in **Figure 4**. Slowly raise the temperature (< 150 °C) until the ferrocene sublimes to the dish cover. The sublimation will proceed slowly. Cooling the dish cover by placing a 400 mL beaker filled with ice water on top of it to facilitate the sublimation [see **Figure 4**]. Remove the glass petri dish cover (bigger size) from the heat, allow it to cool and recover the sublimed ferrocene [Note: This procedure may be repeated twice (or more) until all the purified ferrocene has been obtained.]. Allow the petri dish cover filled with sublimed ferrocene to cool to room temperature. Record the weight(s) of the petri dish covers filled with ferrocene sublimed. Transfer the ferrocene sublimed from the petri dish cover(s) into a clean transparent sample plastic bag.

[Note: The calculation of % yield of the purified ferrocene obtained must be shown in the short lab report.]

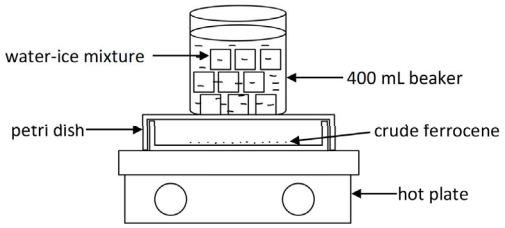


Figure 4. Apparatus for the sublimation of ferrocene

II-b. Characterization of the purified ferrocene

(i) Record of microscopic image of the purified ferrocene

Obtain a microscopic image of your purified ferrocene by using the stereomicroscope *(refer to the stereomicroscope instruction manual in the teaching lab.).* With the aid of GTA or lecturer-in-charge, use the lowest magnification ratio of **14x**, select the best view and record the image of your purified product clearly. Type your matric number on the image captured and send the softcopy of the image to your handphone or NUS email address.

[*Note: The microscopic image obtained must be attached as a part of appendices in the lab report.*]

(ii) Measurement of magnetic susceptibility

Use agate mortar and pestle to grind 0.09-0.12 g of your purified ferrocene into fine powder, and follow the magnetic susceptibility balance instruction manual [Refer to **Appendix**] to measure both volume and mass susceptibilities of your purified ferrocene.

[*Importance: Take pictures to record all the values obtained from the magnetic susceptibility balance in each step stated in the instruction manual (see **Appendix**) and attach them as a part of appendices in the lab report.*]

(iii) UV-visible Spectroscopic Characterization of the purified ferrocene

Prepare the standard solution of purified ferrocene (35-40 mg, weighed accurately by the analytical balance) in methanol in a 25 mL volumetric flask and record its UV-Vis spectrum (300–700 nm). By doing proper dilutions in the fume cupboard, collect the spectra showing weak and strong peaks (A = 0.1–1.0). Determine the molar extinction coefficients ε ($M^{-1}cm^{-1}$) for these peaks obtained from the selected spectra and report them in the proforma lab report.

[*Note: All the ε calculation involved must be handwritten clearly on the selected UV-vis spectra, and ALL the UV-vis spectra obtained must be attached as a part of appendices in the short lab report.*]

(iv) Infrared Characterization of the purified ferrocene

Obtain the solid IR spectrum of the purified ferrocene by *either* ATR-IR *or* KBr disc. Record the spectra within the range 500–4000 cm^{-1}.

[*Note: The IR characterization method chosen must be recorded on your datasheet and the IR spectra obtained must be attached as a part of appendices in the short lab report.*]

(v) ^{1}H-NMR Characterization of the purified ferrocene

Transfer the purified ferrocene (~ 1 mg) into the NMR tube containing 0.6 mL of $CDCl_3$ Label the sample tube clearly and submit it for ^{1}H-NMR.

[Note: ALL the NMR spectra obtained must be attached as a part of appendices in the short lab report.]

(vi) ESI-MS Characterization of the purified ferrocene

Transfer the purified ferrocene (1-2 mg) into a sample tube. Label the sample tube clearly and submit it for Electrospray Ionization Mass Spectrometry (ESI-MS) analysis.

[Note: ALL the ESI-MS spectra obtained must be attached as a part of appendices in the short lab report.]

C. LAB REPORT

Individual **softcopy** short lab report (*via* Canvas-Assignments). Refer to CM3191-inorganic lecture notes for the requirements of short lab report and risk assessment. The "ONE-file" softcopy lab report in pdf format (attached with *the softcopy of the microscopic images, the pictures taken during the measurement of magnetic susceptibility, 1H-NMR, IR, UV-vis and ESI-MS spectra of your products obtained, and your own datasheet(s) for all lab sessions*) must be uploaded into the folder named "Inorganic Expt. 2 short lab report submission" *via* Canvas-Assignments. Hardcopy of lab report and risk assessment will NOT be accepted. Penalty will be applied for any late submission.

D. REFERENCES

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2. (a) G. Wilkinson, M. Rosenblum, M. C. Whiting, R. B. Woodward, *J. Am. Chem. Soc.* **1952**, *74*, 2125–2126. (b) F. A. Cotton, *J. Organomet. Chem.* **2001**, *637-639*, 18–26.
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5. (a) S. Sansook, S. Hassell-Hart, C. Ocasio, J. Spencer, *J. Organomet. Chem.* **2020**, *905*, 121017. (b) N.J. Long, *Metallocenes: An Introduction to Sandwich Complexes*, Blackwell Science Inc., **1998**. (c) Y. Yamaguchi, B.J. Palmer, C. Kutal, T. Wakamatsu, D.B. Yang, *Macromolecules*, **1998**, *31*, 5155. (d) A. Togni, T. Hayashi, *Ferrocenes: Homogeneous Catalysis, Organic Synthesis, Materials Science*, Wiley, **2008**.

E. APPENDICES *(available in Canvas-Files for inorganic experiments)*

Material Safety Data Sheets (MSDSs) of 1,2-dimethoxyethane, acetonitrile, calcium chloride, cyclopentadiene, deionized water, deuterochloroform, dicyclopentadiene, dimethyl sulfoxide, ferrocene, hydrochloric acid 37%, iron(II) chloride tetrahydrate, methanol, potassium bromide, potassium hydroxide

< **Appendix** >

Operation of Magnetic Susceptibility Balance (Model: Sherwood MSB AUTO)



1. Switch on the instrument.
2. Wait until a zero value is obtained (-0.000E-5Z). **Take a picture** on the control unit for such reading.
3. Weight an empty sample tube. Place an empty sample tube of known weight to the detector and obtained a reading.
4. Press the "Tare" function, and observed the display to ensure the numerical value of zero is displayed.
5. Remove the sample tube and introduce the sample. Measure the weight and length of the sample packed, with an analytical balance and a ruler. (Note: The length must be between 15.0 and 25.0 mm.)
6. Press the "Length" function button and observe the display.
   Length = _ _._ mm
7. Enter the length by pressing the appropriate three number keys in order. **Take a picture** on the screen of the control unit before entering the last number.
8. Following the input of the appropriate length, the display indicates "processing" and then returns to the Xv-Tare display. Press the "Weight" function button and observe the display. **Take a picture** on the screen of the control unit before entering the last number.
   Weight = 0._ _ _ _ gm
   (Note: The weight value between 0.0010 grams and 0.9999 grams are expected by the Auto.)
9. The display will return to the Xv-Tare display, press the "Mag Sus" button and observe the display. **Take a picture on the control unit for each of the following readings.**
   (i) Volume Suscept → x.xxx E-xV Tare
   (ii) Mass Suscept → x.xxx E-xM Tare

   [Note: M – Mass susceptibility; V – Volume susceptibility]